

Figure 5: Syntactically right (up) and wrong (down) display filters

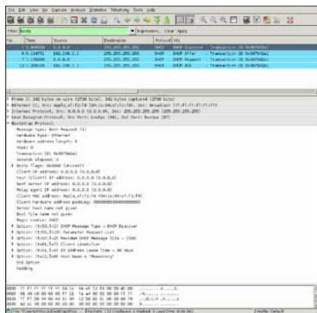


Figure 6: A DHCP transaction

What you can also notice in Figure 5 is that Wireshark is smart enough to understand erroneous IP addresses such as 192.168.257.10. The presented display filter shows only traffic that originates from, or goes to, the 192.168.1.10 IP address.



Note: Display filters do not magically solve problems by themselves. They are extremely useful tools when used the right way, but you still have to interpret the results, find the problem and think about the solutions.

Analysing DHCP traffic

DHCP (Dynamic Host Configuration Protocol) provides configuration information to hosts on TCP/IP networks. It is based on BOOTP (the Bootstrap Protocol) and extends it by adding more capabilities. DHCP and BOOTP both use UDP (ports 67 and 68). DHCP supports, among others, the following basic messages:

- **DHCPDISCOVER:** Clients searching for a DHCP server

send this broadcast message to the LAN, using only the client MAC address, because the client does not have an IP address yet.

- **DHCPDISCOVER:** When a DHCP server receives a **DHCPDISCOVER** message, it responds with a **DHCPDISCOVER** message.
- **DHCPREQUEST:** The **DHCPREQUEST** message provides information to the chosen DHCP server, even if there is only one available.
- **DHCPACK:** This message is the response of the chosen DHCP server. It includes all the required configuration information.

Figure 6 shows a DHCP transaction that involves the airport wireless card of an iMac and an ADSL router that also acts as a DHCP server. Please notice how much easier it is to see the DHCP protocol when using the right Display Filter (**bootp**).

The first packet is the **DHCPDISCOVER** message from the iMac searching for a DHCP server. Since the iMac does not have an IP address yet, the source IP of the packet is 0.0.0.0 and the destination IP is the broadcast IP (255.255.255.255). What distinguishes the airport card of the iMac from the other network devices found in the same LAN is the MAC address of the airport card, which is unique. Therefore, the **DHCPDISCOVER** message should include the MAC address of the device requesting a DHCP server.

The next message is the **DHCPDISCOVER** from the DHCP server with IP 192.168.1.1, and is a broadcast message, since the iMac still has no IP address. Then the iMac requests the DHCP server for the offered configuration parameters with the **DHCPREQUEST** message. Next, the DHCP server sends a **DHCPACK** message back to the iMac that includes the configuration parameters. From now on, the iMac can use the offered configuration information and any parameter that is unique to the iMac, like the IP address, is reserved by the DHCP server and is not offered to any other device.

Wireshark is a powerful tool that every Linux or network administrator should know. I suggest that you first start by learning display filters before going any further. As always, before trying to solve a complex problem, first try to address simpler or smaller issues. 

Web links and bibliography

- [1] Wireshark: <http://www.wireshark.org/>
- [2] Tcpdump and libpcap site: <http://www.tcpdump.org/>
- [3] Internetworking with TCP/IP, Volume I, Douglas E. Comer, Prentice Hall
- [4] DHCP RFC: <http://www.ietf.org/rfc/rfc2131.txt>
- [5] Display Filters Reference: <http://www.wireshark.org/docs/dfref/>
- [6] The TCP protocol RFC: <http://www.ietf.org/rfc/rfc793.txt>
- [7] The UDP protocol RFC: <http://www.ietf.org/rfc/rfc768.txt>

By: Mihalis Tsoukalos

The author enjoys photography, writing articles, programming iOS devices and administering UNIX machines. You can reach him at tsoukalos@sch.gr or @mactsouk.